JURONG JUNIOR COLLEGE
JC2 Preliminary Examination 2018

CANDIDATE NAME

CLASS

BIOLOGY **8876/01**

Paper 1 Multiple Choice **14 September 2018**

1 hour

Additional Materials: Multiple Choice Answer Sheet

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, glue or correction fluid.
Write your name, class and index number on the Answer Sheet in the spaces provided unless this has been done for you.

There are **thirty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.
The use of an approved scientific calculator is expected, where appropriate.

This document consists of **21** printed pages and **1** blank page.

1 A severe inherited condition arises from the failure to produce an enzyme that breaks down glycoproteins in cells. The condition can be diagnosed from an electron micrograph of a patient's cells.

Which abnormality would be observed in these cells?

A an incomplete chromosome due to lack of a gene

B larger lysosomes due to accumulation of glycoprotein

C less endoplasmic reticulum due to a reduction in protein synthesis

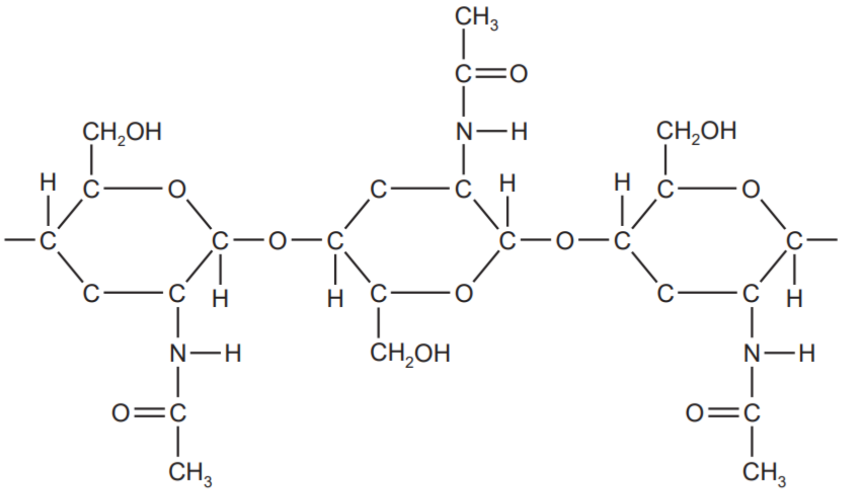
D thinner cell surface membrane due to lack of glycoprotein

2 The table shows a comparison between three features of a prokaryotic cell and a eukaryotic cell.

Which row is correct?

	chromosome structure		ribosome		presence of introns	
	prokaryote	eukaryote	prokaryote	eukaryote	prokaryote	eukaryote
A	circular	linear	80S	70S	rare	common
B	circular	linear	70S	80S	rare	always
C	linear	circular	80S	70S	common	always
D	linear	circular	70S	80S	always	never

3 The diagram shows the structure of the polysaccharide chitin which is found in the cell wall of fungi.



Which statements are correct for chitin and for cellulose?

1 The monomers are joined by 1, 4 glycosidic bonds.

2 Every second monosaccharide in the polysaccharide chain is rotated by 180°.

3 The polysaccharide contains the elements carbon, hydrogen, oxygen and nitrogen.

A 1, 2 and 3

B 1 and 2 only

C 1 and 3 only

D 2 only

4 Which statements about triglycerides and phospholipids are correct?

1 Triglycerides and phospholipids both have a hydrophobic region.

2 Triglycerides are non-polar molecules and phospholipids are polar.

3 Fatty acids in a triglyceride may be saturated or unsaturated but in a phospholipid they are always saturated.

A 1 and 2

B 1 and 3

C 2 only

D 3 only

5 The cells in the roots of beetroot plants contain a red pigment.

When pieces of root tissue are soaked in cold water, some of the red pigment leaks out of the cells into the water.

An experiment was carried out to investigate the effect of temperature on the loss of red pigment from the root cells. It was found that the higher the temperature of the water, the higher the rate of loss of red pigment from the root cells.

Which of these statements could explain this trend?

1 Enzymes in the cells denature as the temperature increases, so the pigment can no longer be used for reactions inside the cells and diffuses out.

2 As the temperature increases, the tertiary structure of protein molecules in the cell surface membrane changes, increasing the permeability of the membrane.

3 Phospholipid molecules gain kinetic energy as temperature rises, increasing the fluidity of the phospholipid bilayer and allowing pigment molecules to diffuse out more easily.

A 1 and 2

B 2 and 3

C 2 only

D 3 only

6 High concentrations of urea break all bonds except covalent bonds in protein molecules. Which level of protein structure would remain unchanged when keratin is treated with urea?

A primary

B secondary

C tertiary

D quaternary

7 Which feature of haemoglobin makes it a globular protein?

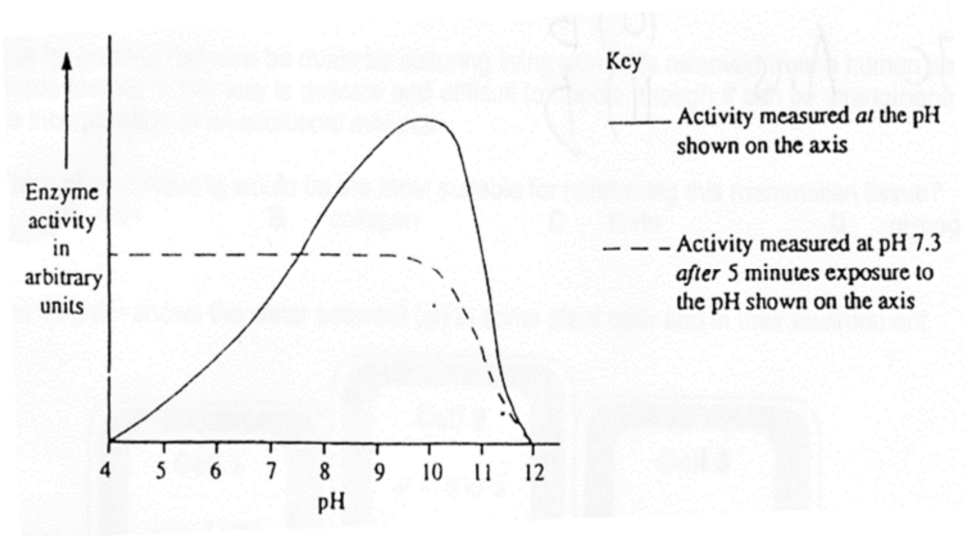
A It has four cross-linked polypeptide chains forming a quaternary structure.

B It has hydrophobic groups on the inside and hydrophilic groups on the outside.

C It has hydrophobic interactions and is insoluble in water.

D It has polypeptide chains which are cross-linked to form sheets.

8 The graph shows the effects of pH on the activity of the enzyme monoamine oxidase, as measured by two different methods.



Which hypothesis is not supported by these results?

A Both acid and strong alkali denature the enzyme.

B The optimum pH is alkaline.

C Strong alkali causes a reversible change in the tertiary structure of the enzyme.

D The change in the catalytic properties of the enzyme caused by acid is reversible.

9 There are two hypotheses to explain how enzymes interact with a substrate.

1 In the lock-and-key hypothesis, enzyme active sites and substrate molecules have complementary shapes.

2 In the induced-fit hypothesis, the substrate causes the active site to become a complementary shape.

What is true only of the induced-fit hypothesis?

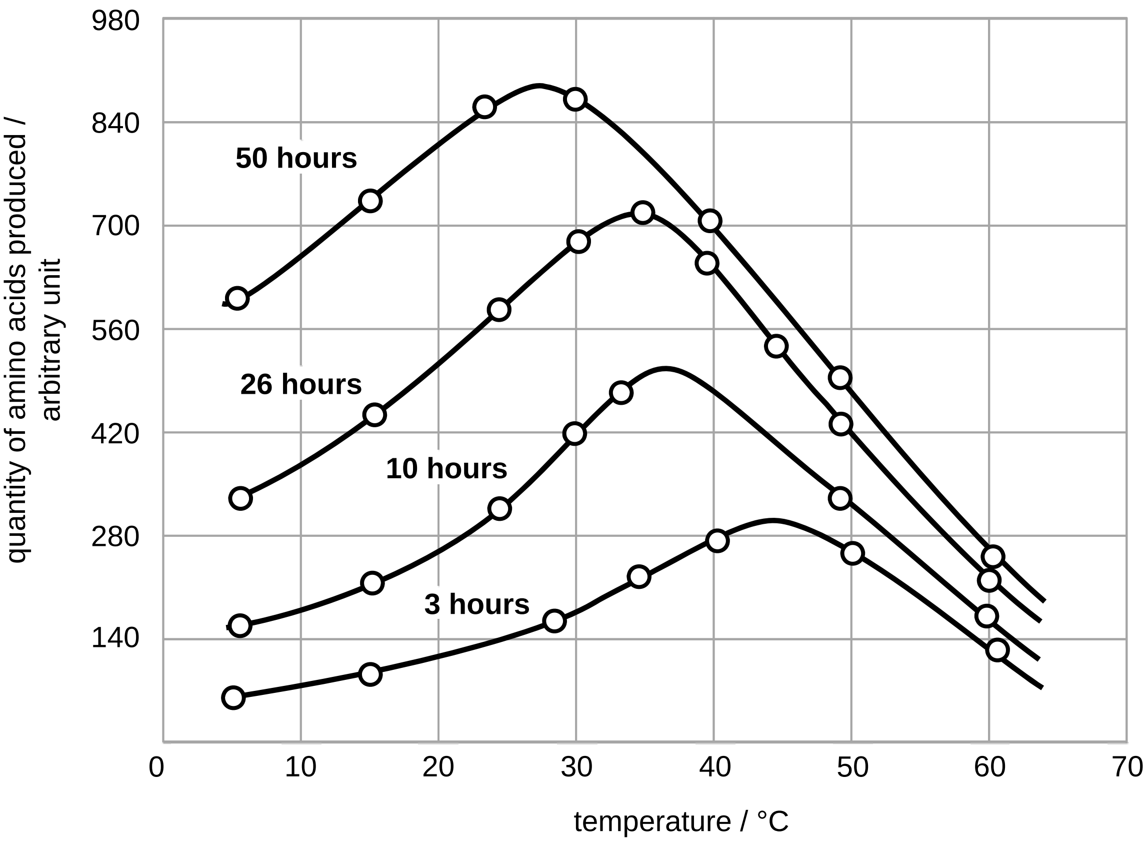
A Adding an excess of substrate does not increase the number of enzyme-substrate complexes.

B An enzyme lowers the activation energy, causing more enzyme-substrate complexes to form.

C An enzyme undergoes a conformational change as it forms enzyme-substrate complexes.

D An increase in enzyme concentration may increase the number of enzyme-substrate complexes.

10 The graph below shows the variation in the effect of a proteolytic enzyme with temperature over time.



Which of the following conditions can be deduced from the graph?

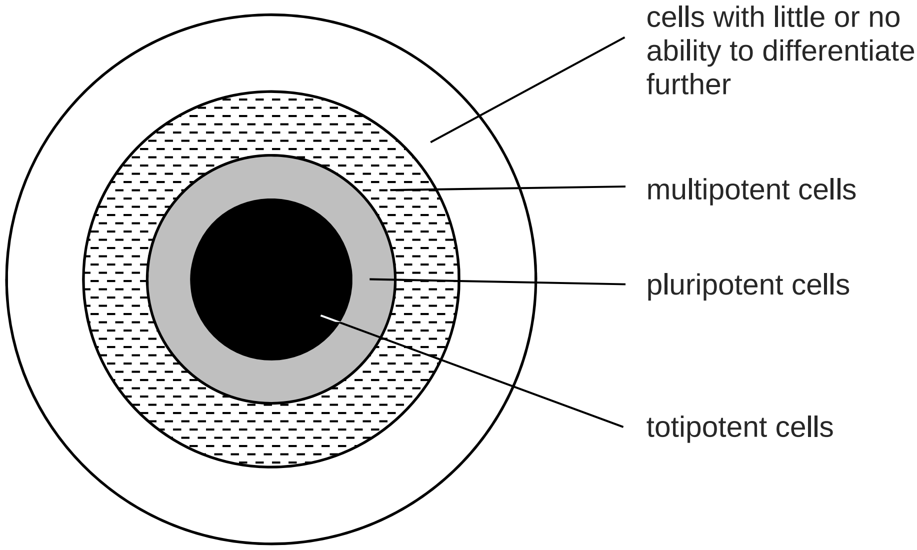
A The optimum temperature for the enzyme is 27 °C.

B The optimum temperature for the enzyme increases as the substrate is hydrolysed.

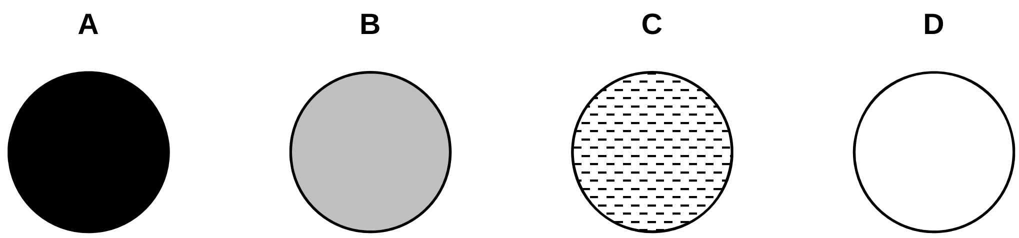
C The optimum temperature increases causing the experiment to last longer.

D The optimum temperature of the enzyme decreases over the duration of the experiment.

11 The diagram represents all human cell types. Each shaded region represents cells with varying degrees of ability to differentiate into different cell types.



Which shaded region includes blood stem cells?



12 Anti-cancer therapy that targets dividing cells can reduce the numbers of the different blood cell types in a person's body.

Which statement about blood stem cells is correct?

A As blood stem cells are totipotent, all the blood cell types can be produced to restore circulating differentiated blood cells to their normal numbers.

B As blood stem cells are undifferentiated, they are unaffected by anti-cancer therapy and will continue to renew themselves to gradually bring numbers of blood cells back to normal.

C Blood stem cells that are collected from a person before anti-cancer therapy and are then put back into the same person after therapy are able to produce new blood cells.

D Blood stem cells removed from the bone marrow, if transplanted to the site where the cancer is located, will differentiate into the cell types of the new location.

13 Which statements concerning DNA and RNA are correct?

1 Adenine and guanine are bases that have a double ring structure; cytosine, thymine and uracil are bases with a single ring structure.

2 An adenine nucleotide from DNA is the same as an adenine nucleotide from RNA; DNA adenine pairs with thymine and RNA adenine pairs with uracil.

3 The base pairing that occurs in a double DNA helix and when RNA is synthesised during transcription is always according to the rule that a purine pairs with a pyrimidine.

4 The two polynucleotides on a DNA molecule run in opposite directions so that the double helix formed has two strands that are parallel to each other.

A 1, 2 and 3

B 1, 2 and 4

C 1, 3 and 4

D 2, 3 and 4

14 Which row shows the correct functions of the three different proteins involved in DNA replication?

	helicase	single-strand binding protein	DNA polymerase
A	adds DNA nucleotides to the 3’ end of a growing polynucleotide strand	make strands available as templates	prevents original strands reforming complementary base pairs
B	prevents original strands reforming complementary base pairs	adds DNA nucleotides to the 3’ end of a growing polynucleotide strand	make strands available as templates
C	prevents original strands reforming complementary base pairs	make strands available as templates	adds DNA nucleotides to the 3’ end of a growing polynucleotide strand
D	make strands available as templates	prevents original strands reforming complementary base pairs	adds DNA nucleotides to the 3’ end of a growing polynucleotide strand

15 The table compares the structure and function of some elements involved in transcription.

	made of protein	interacts with protein	codes for protein
RNA polymerase	1	2	3
promoter	4	5	6
terminator	7	8	9
gene	10	11	12

Which combination of numbers link the four elements listed to their structures and functions?

A 1, 5, 6, 9 and 12

B 1, 5, 8, 11 and 12

C 2, 6, 7, 8 and 11

D 3, 4, 8, 10 and 12

16 Two enzymes, X and Y, are each encoded by different alleles of the same gene. The amino acid sequences of the two enzymes differ between positions 87 and 91 of the polypeptide.

The amino acid sequences of enzymes X and Y, and the corresponding DNA sequence of enzyme X from position 86 to position 93 of the polypeptides, are shown in the table below.

	←N terminal end amino acid position C terminal end→							
	86	87	88	89	90	91	92	93
DNA triplet codes for enzyme X	TTT	TCA	GGT	AGT	GAA	TTA	CGA	CGA
amino acid sequence of enzyme X	lys	ser	pro	ser	leu	asn	ala	ala
amino acid sequence of enzyme Y	lys	val	his	his	leu	met	ala	ala

The actual mRNA codons for the amino acids in these positions for enzymes X and Y, are shown in the table below.

amino acid	Lys	Ser	Pro	Leu	Asn	Ala	Val	His	Met
mRNA codon(s)	AAA	AGU UCA	CCA	CUU UUA	AAU	GCU	GUC	CAU CAC	AUG

What could account for the difference in amino acid sequence of enzymes X and Y?

A A single frame shift by deletion in the DNA code at position 87.

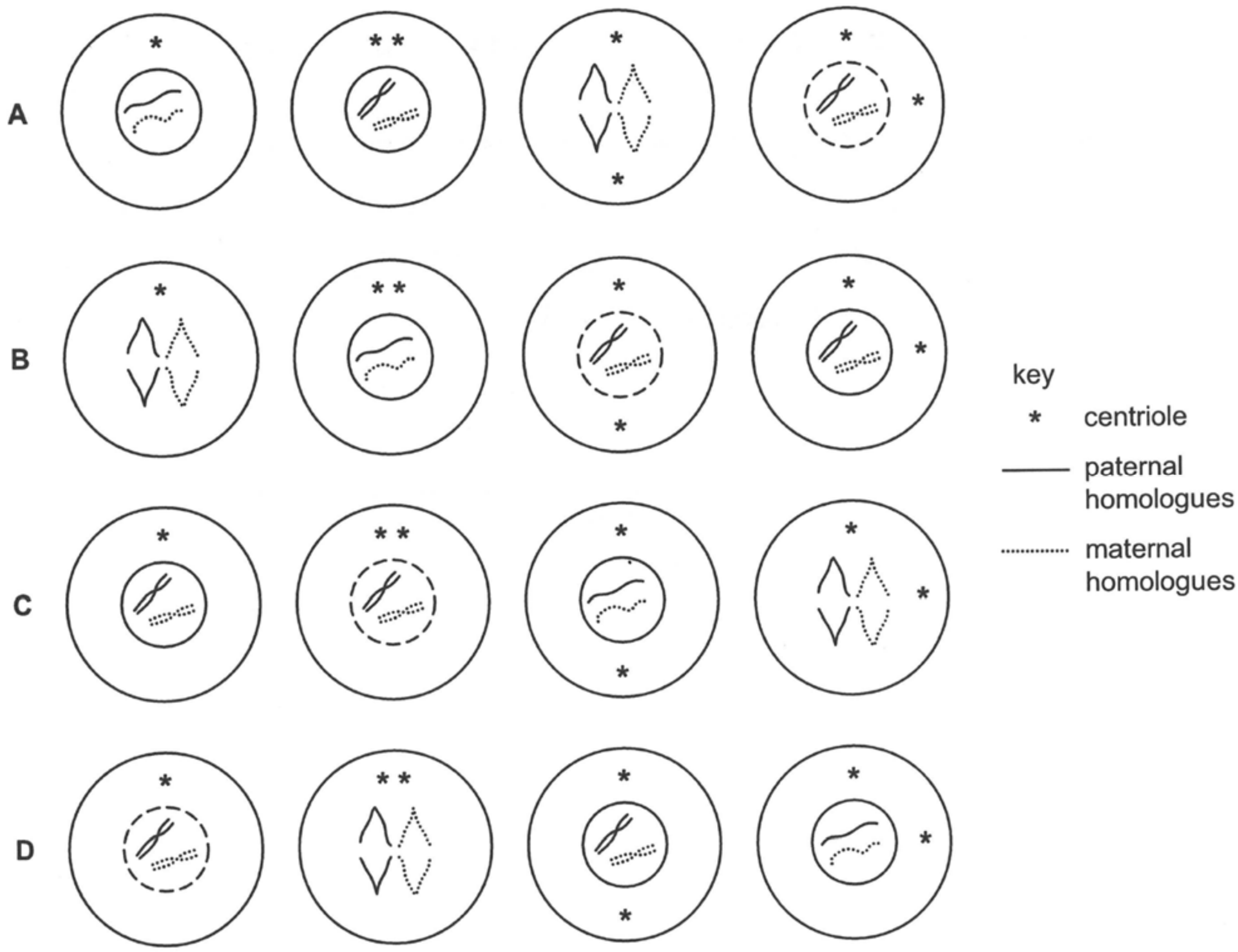
B Frame shift mutations in the DNA codes at position 87 and position 90.

C A change in the sequences of the second and third nucleotides at positions 87 and 88 of the DNA codes and frameshifts at positions 89 and 91

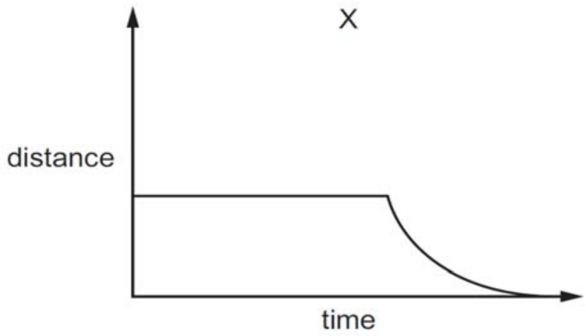
D A deletion in the DNA code at position 87 and an insertion into the DNA code at position 92.

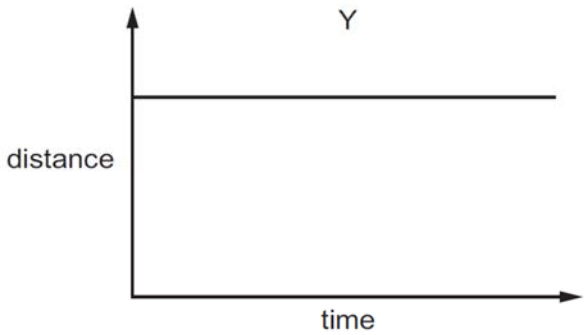
17 The diagram shows cells in the mitotic cell cycle.

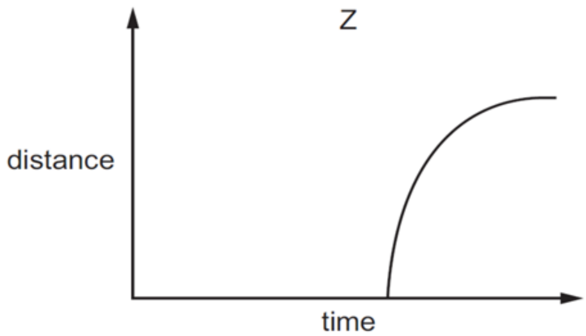
In which row does the appearance of the chromosomes match the appearance of the nuclear envelopes and the positions of the centrioles in all four cells?



18 The graphs show various distance measurements taken from metaphase of mitosis onwards. The graphs are to scale when compared to one another.







Which row correctly identifies the distance measurement for each graph?

	X	Y	Z
A	distance between poles of spindle	distance between sister chromatids	distance of centromeres from poles of spindle
B	distance between poles of spindle	distance of centromeres from poles of spindle	distance between sister chromatids
C	distance of centromeres from poles of spindle	distance between poles of spindle	distance between sister chromatids
D	distance of centromeres from poles of spindle	distance between sister chromatids	distance between poles of spindle

19 It has been suggested that breast cancer cells produce high levels of hydrogen peroxide. This causes connective tissue cells near the cancer cells to digest some of their mitochondria, releasing nutrients which feed the cancer cells.

Which observations made on breast cancer cells and connective tissue cells growing in tissue culture support this view?

1 Breast cancer cells grown alone produce hydrogen peroxide.

2 Treating breast cancer cells with hydrogen peroxide causes apoptosis.

3 Connective tissue cells grown with breast cancer cells have reduced mitochondrial activity.

4 Treating breast cancer cells with peroxidases increases cancer cell death.

A 1, 2, 3 and 4

B 1, 3 and 4 only

C 1 and 4 only

D 2 and 3 only

20 Which are correct statements about the need for a reduction division?

1 Reduction division needs to occur so that a parent will contribute to the zygote one set of chromosomes that are similar in size and number to the set contributed by the other parent and contain the same genes, but not necessarily the same alleles.

2 There is a requirement to reduce the diploid number of chromosomes, characteristic of the species, to produce cells with a haploid number in preparation for fertilisation and the subsequent restoration of the original diploid number.

3 Without reduction division occurring, the fusion of two gametes, each with a diploid number, would give rise to a chromosome aberration and cells that would have more than two copies of any one gene.

A 1, 2 and 3

B 1 and 2 only

C 1 and 3 only

D 2 and 3 only

21 In a monohybrid, sex-linked genetic cross involving dominant and recessive alleles, different phenotypes are observed.

Which statement correctly explains these different phenotypes?

A Expression of the recessive allele only occurs in males, because the Y chromosome lacks the relevant gene.

B The nucleotide sequences of the two alleles each produces a different mRNA molecule, only one of which is translated into a functional protein.

C The recessive allele present in the male is unlikely to be transcribed and translated, as the male does not have a corresponding nucleotide sequence on the shorter Y chromosome.

D Transcription of alleles at different loci leads to transcription and translation of active and inactive enzymes.

22 The feather colour of a certain breed of chicken is controlled by codominant alleles. A cross between a homozygous black-feathered chicken and a homozygous white-feathered chicken produces all speckled chickens. What phenotypic ratios would be expected from a cross between two speckled chickens?

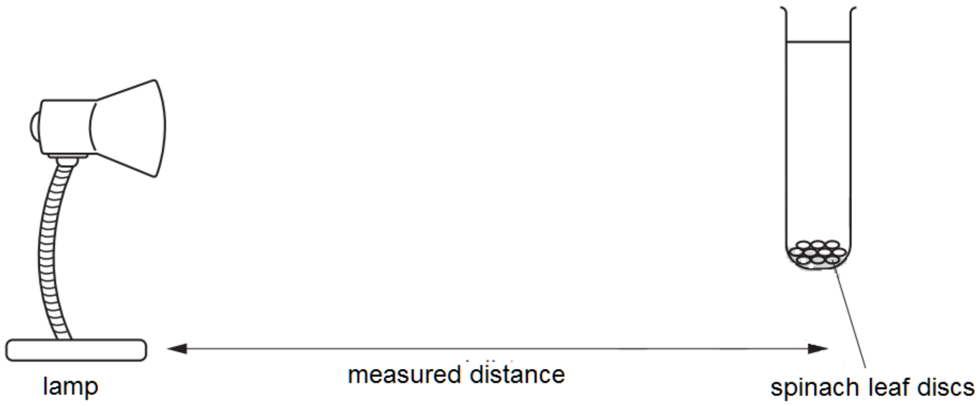
A all speckled

B 1 black feathers : 1 white feathers

C speckled, black feathers and white feathers in equal numbers

D 1 black feathers : 2 speckled feathers : 1 white feathers

23 A student carried out an investigation into the effect of light intensity on photosynthesis. Several groups of spinach leaf discs were placed in test tubes of water. The discs all sank to the bottoms of the tubes. Each tube was placed at a measured distance from a lamp.



As photosynthesis occurs, the build-up of oxygen gas in the leaf discs causes them to rise from the bottom of the tube upwards. The results are shown in the table below.

tube number	distance from lamp / mm	time taken for five discs to float / s
1	50	125
2	100	210
3	150	360
4	200	600
5	250	none floated in the time available

Which of these statements are true?

1 The compensation point occurs between 200 and 250 mm.

2 A variable which is controlled is the distance of the tube from the light source.

3 The time taken for the discs to rise is directly proportional to the distance from the lamp.

A 1, 2 and 3

B 1 and 2 only

C 1 only

D 2 and 3 only

24 What happens in both respiration and photosynthesis?

1 ADP is phosphorylated.

2 ATP is hydrolysed.

3 Electrons pass through ATP synthase.

4 NADP is reduced.

5 Protons pass through ATP synthase.

6 Triose phosphates are decarboxylated.

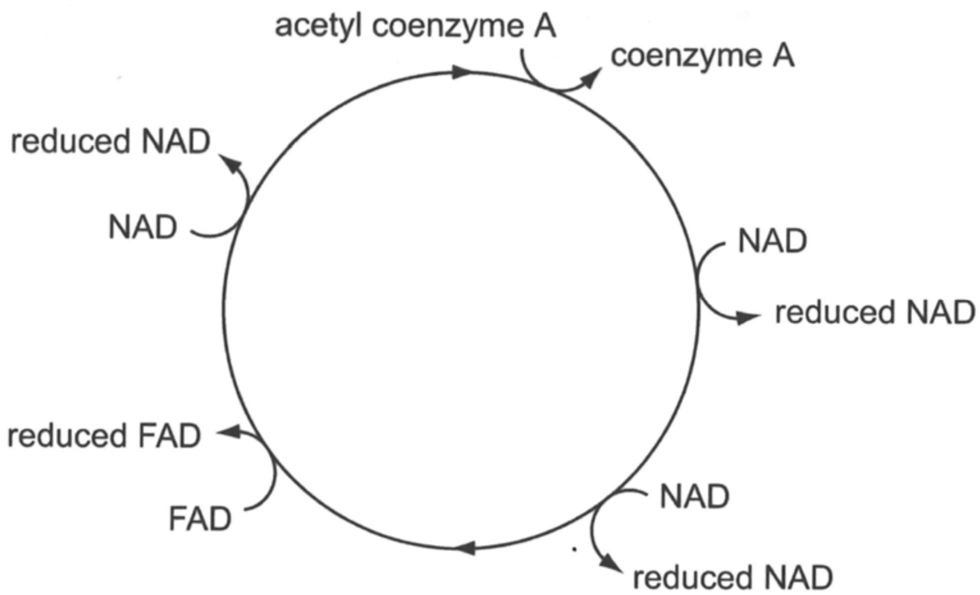
A 1, 4 and 5

B 1 and 5 only

C 2, 4 and 6

D 3 and 6

25 The diagram shows the reactions of the hydrogen carriers in the Krebs cycle.



The average yield of ATP, in oxidative phosphorylation, is 2.5 molecules from each molecule of reduced NAD and 1.5 molecules from each molecule of reduced FAD.

What is the average yield of ATP from the hydrogen carriers reduced in the Krebs cycle from one molecule of glucose?

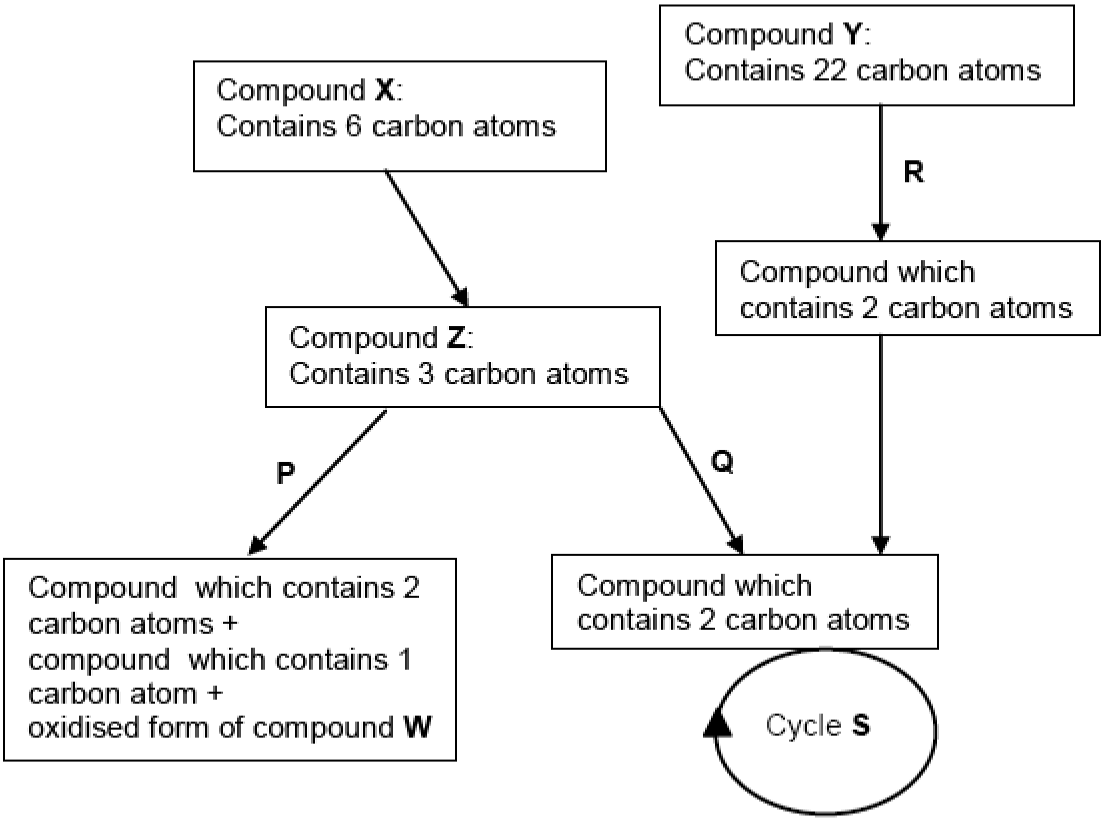
A 9

B 18

C 28

D 32

26 The diagram shows the flow of carbon atoms in cellular respiration in a plant cell, where processes **P**, **Q** and **R** are key stages.



Which statements are correct?

1 Process P involves the formation of lactic acid and regeneration of the oxidised form of compound W.

2 Compound Y is starch which hydrolyses into acetyl-CoA.

3 In process Q, compound Z undergoes oxidative decarboxylation.

4 Cycle S produces large amounts of reduced coenzymes for oxidative phosphorylation.

4 One molecule of compound X yields 36-38 ATP molecules when completely oxidised.

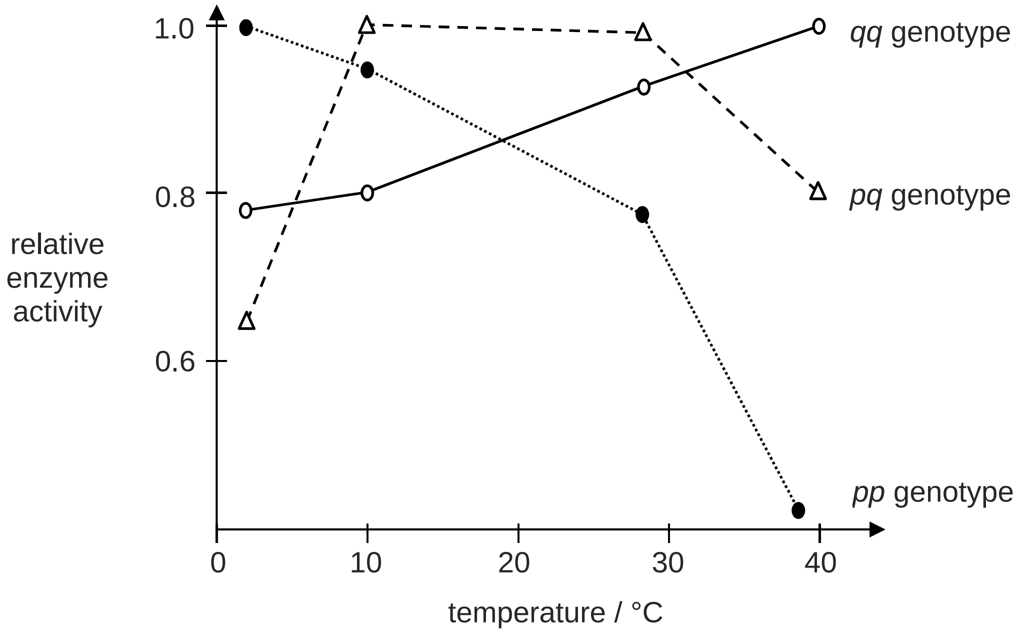
A 1, 2 and 4

B 1 and 3

C 2 and 5

D 3, 4 and 5

27 In the North America catfish *Catostomus clarki*, two alleles, represented by *p* and *q*, control the synthesis of a vital enzyme. The three possible genotypes (*pp*, *pq*, *qq*) lead to the synthesis of variations of the same enzyme with different temperature optima as shown in the graph below.



When the mean annual temperature is 5 °C, which is correct?

A Allele *p* will be positively selected for.

B Allele *p* will become dominant and *q* recessive.

C The heterozygotes will have a selective advantage over the homozygotes.

D No genotype will have a selective advantage over another.

28 Animals with horizontal stripes are bitten less frequently by tsetse flies. The flies carry diseases that infect zebras.

Which explains how zebras might evolve to have more horizontal stripes?

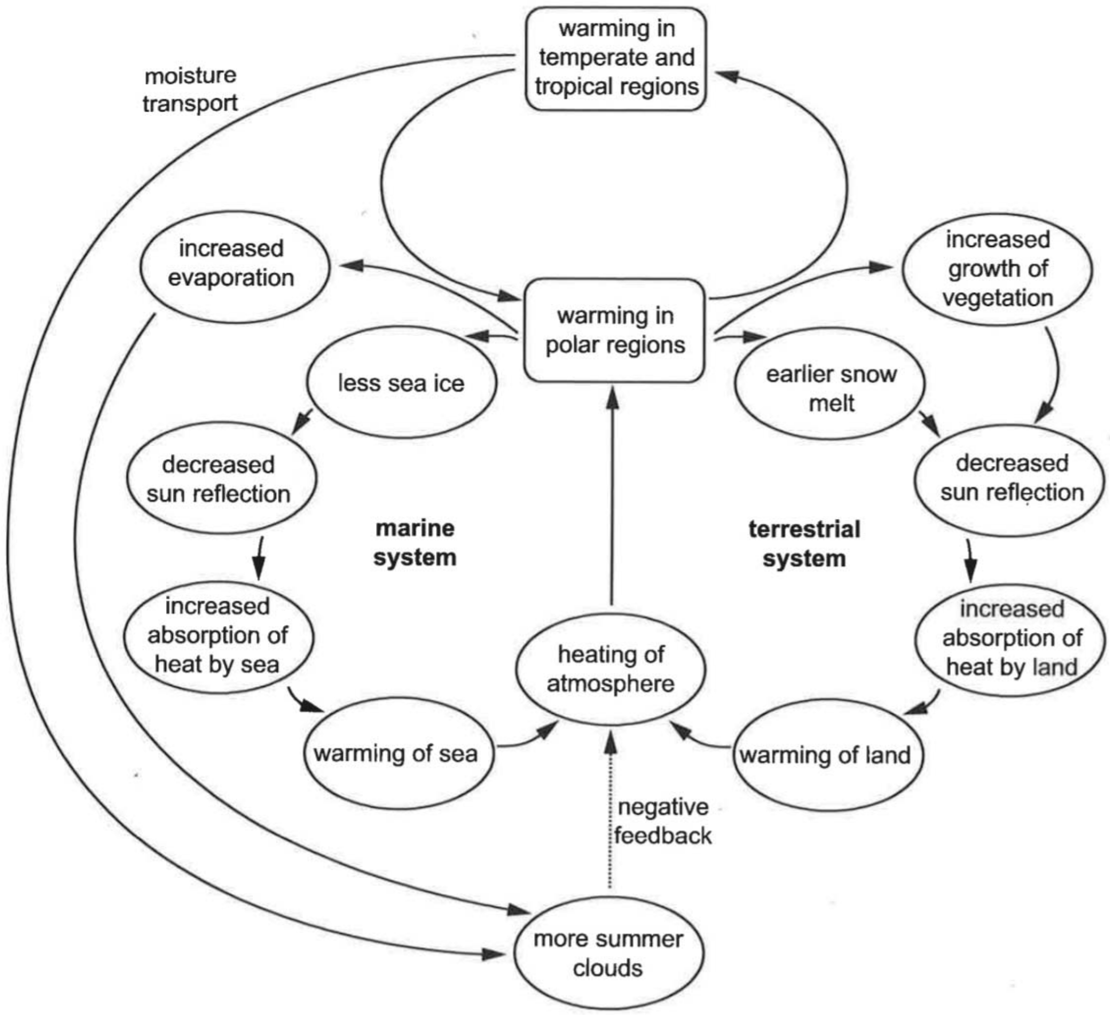
A Bites from tsetse flies cause mutations. If a zebra has a mutation it will die and not pass its genes to its offspring which will not have more horizontal stripes.

B If two zebras with horizontal stripes mate, their offspring will have more horizontal stripes. Horizontal stripes will become dominant. This is natural selection.

C Tsetse flies are a selection pressure. The zebras would gradually develop more horizontal stripes and pass them on to their offspring so they are not bitten by flies.

D Zebras with more horizontal stripes get fewer diseases from tsetse flies. These zebras live longer and breed more, passing the allele for more horizontal stripes to their offspring.

29 The diagram shows the effect of increasing temperatures on the ice and snow cover at the polar regions.



Which effect of higher temperatures in the polar regions could increase global warming?

A Increased evaporation leads to more rainfall, which absorbs heat from the land and the sea.

B Melting of ice and snow results in less reflection of sunlight and more heat absorption by the Earth.

C Melting of sea ice caused more cloud formation, which increases absorption of heat in the atmosphere.

D Earlier melting of snow allows vegetation cover to increase faster, reducing loss of heat from the surface of the Earth.

30 Increases in global temperatures have been linked in some species to changes in their distribution or to the timing of stages in their life-cycles. Examples in North America include:

- earlier nesting of several species of birds
- earlier migration of birds from southern winter habitats to northern breeding habitats
- movement of warm water fish into rivers inhabited by cold water species
- an increase in the range of boreal (coniferous) forest as it moves north and invades tundra grassland.

If global temperatures continue to increase, which consequences are likely to occur in North America as a result of these changes?

1 mismatch between the timing of breeding seasons and maximum food availability

2 decrease in populations of all species

3 reduction in the area of tundra grassland

4 decrease in available habitat for cold water species of fish

A 1, 2, 3 and 4

B 1, 3 and 4 only

C 1 and 2 only

D 3 and 4 only

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